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College of Science and Technology
Rinchending: Bhutan

DBS101
Database Systems Fundamentals
(SS2026)

Practical 1 Report

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RUB Wheel of Academic Law: Academic Dishonesty

Section H2 of the Royal University of Bhutan's *Wheel of Academic Law* provides the following definition of academic dishonesty:

Academic dishonesty may be defined as any attempt by a student to gain an unfair advantage in any assessment. It may be demonstrated by one of the following:

1. **Collusion:** the representation of a piece of unauthorized group work as the work of a single candidate.
2. **Commissioning:** submitting an assignment done by another person as the student's own work.
3. **Duplication**: the inclusion in coursework of material identical or substantially similar to material which has already been submitted for any other assessment within the University.
4. **False declaration**: making a false declaration to receive special consideration by an Examination Board or to obtain extensions to deadlines or exemption from work.
5. **Falsification of data**: presentation of data in laboratory reports, projects, etc., based on work purported to have been carried out by the student, which has been invented, altered or copied by the student.
6. **Plagiarism**: the unacknowledged use of another's work as if it were one's own.

Examples are:

- verbatim copying of another's work without acknowledgement.
- paraphrasing of another's work by simply changing a few words or altering the order of presentation, without acknowledgement.
- ideas or intellectual data in any form presented as one's own without acknowledging the source(s).
- making significant use of unattributed digital images such as graphs, tables, photographs, etc., taken from test books, articles, films, plays, handouts, the internet, or any other source, whether published or unpublished.
- submission of a piece of work which has previously been assessed for a different award or module or at a different institution as if it were new work.
- use of any material without prior permission of copyright from the appropriate authority or owner of the materials used".

1.Aim

The purpose of this project is to build a simple web version of a social media platform like TikTok. Using Next.js and Tailwind CSS, it will allow users to watch videos, explore profiles, and interact with content, giving hands-on experience in creating modern and interactive web applications.

2.Theory

A Database Management System (DBMS) is software that helps store, organize, and manage data efficiently. SQL (Structured Query Language) is used to interact with the database to add, view, update, or remove data.

In this lab, we used basic SQL commands: CREATE DATABASE to make a new database, CREATE TABLE to define tables, INSERT INTO to add data, SELECT to view data, UPDATE and DELETE to modify records, and JOIN to combine related tables.

We created two tables, department and student, linked by a foreign key (dept_id) to connect students to their departments.

3. Procedure / Steps

1- Create the department table

```
    -> ^C
mysql> CREATE TABLE department (
    ->     dept_id INT PRIMARY KEY,
    ->     dept_name VARCHAR(50) NOT NULL,
    ->     hod_name VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.05 sec)
```

2-Create the student table

```
mysql> CREATE TABLE student (
    ->     student_id INT PRIMARY KEY,
    ->     student_name VARCHAR(50) NOT NULL,
    ->     age INT,
    ->     city VARCHAR(50),
    ->     dept_id INT,
    ->     FOREIGN KEY (dept_id) REFERENCES department(dept_id)
    -> );
Query OK, 0 rows affected (0.09 sec)
```

3-Insert records into department

```
mysql> INSERT INTO department (dept_id, dept_name, hod_name) VALUES
    -> (1, 'Information Technology', 'Sonam Choden'),
    -> (2, 'Business Management', 'Pema Dorji'),
    -> (3, 'Civil Engineering', 'Kinley Wangmo');
Query OK, 3 rows affected (0.02 sec)
Records: 3  Duplicates: 0  Warnings: 0
```

4- Insert records into student

```
mysql> INSERT INTO student (student_id, student_name, age, city, dept_id) VA
LUES
    -> (101, 'Tashi Dorji', 20, 'Thimphu', 1),
    -> (102, 'Pema Lham', 21, 'Paro', 2),
    -> (103, 'Sonam Wangchuk', 19, 'Punakha', 1),
    -> (104, 'Dechen Yangzom', 22, 'Gelephu', 3),
    -> (105, 'Pelden', 20, 'Samdrup Jongkhar', 2);  /* Changed from Karma Ts
hering to Pelden */
Query OK, 5 rows affected (0.02 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

5- Display all records from both tables

```
mysql> -- Display all department records
mysql> SELECT * FROM department;
+---------+-----------------------+---------------+
| dept_id | dept_name             | hod_name      |
+---------+-----------------------+---------------+
|       1 | Information Technology | Sonam Choden  |
|       2 | Business Management    | Pema Dorji    |
|       3 | Civil Engineering      | Kinley Wangmo |
+---------+-----------------------+---------------+
3 rows in set (0.00 sec)
```

6- Show student names along with their departmenturur names

```
mysql> SELECT s.student_name, d.dept_name
    -> FROM student s
    -> INNER JOIN department d ON s.dept_id = d.dept_id;
+----------------+-----------------------+
| student_name   | dept_name             |
+----------------+-----------------------+
| Tashi Dorji    | Information Technology |
| Sonam Wangchuk | Information Technology |
| Pema Lham      | Business Management    |
| Pelden         | Business Management    |
| Dechen Yangzom | Civil Engineering      |
+----------------+-----------------------+
5 rows in set (0.00 sec)
```

7- Display students whose age is greater than 20

```
mysql> SELECT * FROM student
    -> WHERE age > 20;
+------------+----------------+------+---------+---------+
| student_id | student_name   | age  | city    | dept_id |
+------------+----------------+------+---------+---------+
        102 | Pema Lham      |   21 | Paro    |       2 |
        104 | Dechen Yangzom |   22 | Gelephu |       3 |
+------------+----------------+------+---------+---------+
2 rows in set (0.00 sec)
```

8- Display students in the Information Technology department

```
mysql> SELECT s.*
    -> FROM student s
    -> INNER JOIN department d ON s.dept_id = d.dept_id
    -> WHERE d.dept_name = 'Information Technology';
+------------+----------------+------+---------+---------+
| student_id | student_name   | age  | city    | dept_id |
+------------+----------------+------+---------+---------+
|        101 | Tashi Dorji    |   20 | Thimphu |       1 |
|        103 | Sonam Wangchuk |   19 | Punakha |       1 |
+------------+----------------+------+---------+---------+
2 rows in set (0.00 sec)
```

9- Update the city of Tashi Dorji to Wangdue

```
mysql> UPDATE student
    -> SET city = 'Wangdue'
    -> WHERE student_name = 'Tashi Dorji';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE student
    -> SET age = age + 1
    -> WHERE student_name = 'Pema Lham';
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student WHERE student_name = 'Pema Lham';
+------------+--------------+------+------+---------+
| student_id | student_name | age  | city | dept_id |
+------------+--------------+------+------+---------+
|        102 | Pema Lham    |   22 | Paro |       2 |
+------------+--------------+------+------+---------+
1 row in set (0.00 sec)
```

10- Increase the age of Pema Lham by 1

```
mysql> SHOW TABLES;
+----------------------+
| Tables_in_college_db |
+----------------------+
| department           |
| student              |
+----------------------+
2 rows in set (0.02 sec)

mysql>
```

11- Delete the record of Karma Tshering

```
mysql> SHOW TABLES;
+----------------------+
| Tables_in_college_db |
+----------------------+
| department           |
| student              |
+----------------------+
2 rows in set (0.00 sec)

mysql> DESCRIBE department;
+-----------+-------------+------+-----+---------+-------+
| Field     | Type        | Null | Key | Default | Extra |
+-----------+-------------+------+-----+---------+-------+
| dept_id   | int         | NO   | PRI | NULL    |       |
| dept_name | varchar(50) | NO   |     | NULL    |       |
| hod_name  | varchar(50) | YES  |     | NULL    |       |
+-----------+-------------+------+-----+---------+-------+
3 rows in set (0.02 sec)

mysql>
```

]

12-Display the final contents of the student table

```
mysql> SELECT * FROM student WHERE student_name = 'Pelden';
Empty set (0.00 sec)

mysql> SELECT * FROM student;
+------------+----------------+------+---------+---------+
| student_id | student_name   | age  | city    | dept_id |
+------------+----------------+------+---------+---------+
|        101 | Tashi Dorji    |   20 | Wangdue |       1 |
|        102 | Pema Lham      |   22 | Paro    |       2 |
|        103 | Sonam Wangchuk |   19 | Punakha |       1 |
|        104 | Dechen Yangzom |   22 | Gelephu |       3 |
+------------+----------------+------+---------+---------+
4 rows in set (0.00 sec)
```

13- Count how many students are in each department

```
mysql> SELECT d.dept_name, COUNT(s.student_id) AS student_count
    -> FROM department d
    -> LEFT JOIN student s ON d.dept_id = s.dept_id
    -> GROUP BY d.dept_id, d.dept_name;
+------------------------+---------------+
| dept_name              | student_count |
+------------------------+---------------+
| Information Technology |             2 |
| Business Management    |             1 |
| Civil Engineering      |             1 |
+------------------------+---------------+
3 rows in set (0.00 sec)
```

14- Display student name, city, department name, and HOD name

```
mysql> SELECT s.student_name, s.city, d.dept_name, d.hod_name
    -> FROM student s
    -> INNER JOIN department d ON s.dept_id = d.dept_id;
+----------------+---------+------------------------+---------------+
| student_name   | city    | dept_name              | hod_name      |
+----------------+---------+------------------------+---------------+
| Tashi Dorji    | Wangdue | Information Technology | Sonam Choden  |
| Sonam Wangchuk | Punakha | Information Technology | Sonam Choden  |
| Pema Lham      | Paro    | Business Management    | Pema Dorji    |
| Dechen Yangzom | Gelephu | Civil Engineering      | Kinley Wangmo |
+----------------+---------+------------------------+---------------+
4 rows in set (0.00 sec)

mysql>
```

15- Show only departments that have more than 1 student

```
mysql> SELECT d.dept_name, COUNT(s.student_id) AS student_count
    -> FROM department d
    -> INNER JOIN student s ON d.dept_id = s.dept_id
    -> GROUP BY d.dept_id, d.dept_name
    -> HAVING COUNT(s.student_id) > 1;
+------------------------+---------------+
| dept_name              | student_count |
+------------------------+---------------+
| Information Technology |             2 |
+------------------------+---------------+
1 row in set (0.00 sec)

mysql>
```

Conclusion

In this lab, we learned how to create and manage a database using SQL commands. We practiced creating tables with primary and foreign key constraints, inserting data, and retrieving information using different SQL queries.

This lab helped us understand how relational databases work and how multiple tables can be connected using keys. It also improved our practical skills in executing SQL queries for data retrieval, updating records, and managing database information effectively.

References

1. Korth, H. F., Silberschatz, A., & Sudarshan, S. (2020). *Database System Concepts* (7th Edition). McGraw-Hill Education.
2. Elmasri, R., & Navathe, S. B. (2017). *Fundamentals of Database Systems* (7th Edition). Pearson.